

Armaan Patel
Dr. Gupta
CS 453 Senior Coordinating Seminar
04 May 2026

Capstone Project Final Report

I worked on the creation of an AI Enterprise Risk Management Tool for my capstone project. It is an end-to-end system for analyzing and forecasting operational risk using real-world data. The main objective of the project was to step outside of typical analysis and reporting based on data. I wanted to predict future risk on an organizational level. By using the P-COLD (Public Collection of Operational Loss Data) Chinese operational loss dataset, I was able to integrate data engineering, machine learning, and business intelligence to create an interactive dashboard that supports proactive risk management. The system is structured in three main steps: data engineering, modeling, and visualization.

The project began with the creation of a fake dataset that I created based on the Basel framework for operational risk. However, after a recommendation from Dr. Gupta, I switched to using the P=COLD, which would give my project real-world data credibility and would also allow me to get more accurate predictive models for real risk events.

I had to clean up the P-COLD dataset and convert fields to accommodate the dashboard scope. Through feature engineering, I was able to capture risk patterns, lag features for specific months, and rolling aggregates for specific months. Additional features include loss severity indicators, frequency-based encodings of categorical variables, and derived metrics.

The cleaned dataset was structured into a star schema, which includes a central fact table and multiple dimension tables. This design supported efficient filtering and slicing of risk data in the dashboard.

The main core of the project was the creation of a predictive model that estimates the probability of a operation risk event occurring within a future time window. The primary forecasting model was 90 days, with additional forecasting models for 180-day and 365-day horizons to further analyze its forecasting capabilities.

Three models were tested: logistic regression, random forest, and extra trees. Logistics regression was chosen as a baseline model, random forest as the primary model, and extra trees as an alternative ensemble model. I decided not to use more than three models, as that would clutter the dashboard and would require me to spend more time learning about new models and how to implement them.

Threshold tuning was applied due to high-loss rarity to optimize the model based on the F1 score. Model performance was evaluated based on ROC=AUC, precision, recall, and F1 score. These performance indicators were chosen to give a clear and industry-standard-based model performance analysis.

A Power BI dashboard was the final piece of my project. It consisted of five pages:

- Executive Overview - summarizes key metrics from the P-COLD dataset, such as total losses, event counts, and model-predicted risk score.
- Historical Operational Risk Trends - visualizes the evolution of risks.
- Risk Concentration and Severity - identification of high-impact arrears across business units and event types.
- AI Forecasting - highlights high-risk areas across business units, event types, provinces, and causal factors.
- Model Performance - compares models and evaluates performance across 90-day, 180-day, and 365-day horizons.

The multipage dashboard allows users to view historical and forecasted data, allowing for future-looking analysis.

From the dashboard analysis, a few things can be seen. Internal fraud is the most frequent operational risk category. Only a small number of high-severity events account for a large share of total losses. Risk is concentrated in specific business lines and event types. Predictive performance of models improved when using longer forecasting horizons. The Extra Trees model consistently outperformed the other models across evaluation metrics. Through these findings, we can see that operational risk is not evenly spread across and predictive modeling can help find areas that need proactive attention to avoid future risk and loss.

One of the main challenges was finding a real-world dataset. At the start of the project, finding a dataset that was real data that was based on a structured format was difficult, since it's mostly proprietary, and not many companies are willing to showcase their flaws or risks to the public. After a long research period, I was able to find the P-COLD dataset, which is for this exact purpose of analyzing risk and finding ways to combat it.

Another challenge was learning how to properly fine-tune the models due to class imbalance. Since critical and high-loss events are fewer in comparison to medium and low events. To address this, learning how to threshold tune and changing evaluation metrics were done to better accommodate the models for the class imbalance.

The last challenge was integrating the data pipeline into Power BI and continuous debugging and refinement due to column name and structuring issues. This process took a while as I had to dive deeper into DAX queries for Power BI, then later, learn the integration of the dashboard into my portfolio for easy access for everyone to view.

My capstone project demonstrates how data science and analytics can be combined to build an operational risk management system. Integrating predictive modeling with a structured data pipeline into an interactive dashboard allows the system to showcase historical insights and future-looking risk. The dashboard highlights past risk patterns and enables proactive decision-making. This methodology is to reflect how organizations can leverage data and AI to enhance enterprise risk management.